

# Morning Paper Report - Tuesday, August 04, 2026

Recent, non-repeated papers matched to visual working memory, visual search, modeling, working-memory representation, neural-network memory models, AI for human memory, and egocentric predictive-coding memory themes.

## 1. Transformer Mechanisms Mimic Frontostriatal Gating Operations When Trained on Human Working Memory Tasks

Aaron Traylor, Jack Merullo, Michael J. Frank, Ellie Pavlick  
Annual Meeting of the Cognitive Science Society, 2024

Source: Semantic Scholar; matched terms: working memory, computational model, computational models, neural network, neural networks, transformer

Link: <https://www.semanticscholar.org/paper/8f48f5b22fe0e71227d002fac0bff1ee937c45bc>

**Goal:** Working memory (WM) is thought to rely on sophisticated frontostriatal mechanisms for selective gating, supporting updating and readout of information to and from distinct memory "addresses" (neural populations)

**Method:** According to computational models, capacity limitations in WM arise due to challenges in "role addressability": learning to correctly bind items in memory to their respective roles and assigning credit to the corresponding gating operations

**Results:** Moreover, these gating strategies support enhanced generalization and variable binding and increase the models' effective capacity to store and access multiple items in memory, resembling the constraints found in frontostriatal models

**Conclusion:** Working memory (WM) is thought to rely on sophisticated frontostriatal mechanisms for selective gating, supporting updating and readout of information to and from distinct memory "addresses" (neural populations)

## 2. Alterations in Neural Networks During Working Memory Encoding Related to Cognitive Impairment in Temporal Lobe Epilepsy

Liping Pan, Yakun Wu, J. Bao, Dandan Guo, Xin Zhang, Jiajing Wang, et al.  
Frontiers in Human Neuroscience, 2022

Source: Semantic Scholar; matched terms: working memory, neural network, neural networks

Link: <https://www.semanticscholar.org/paper/4f2052e0567c4b25b157aeee3bd29e2a81f317ce>

**Goal:** Objective: The aim of the current study was to investigate the alterations in the neural networks of patients with temporal lobe epilepsy (TLE) during working memory (WM) encoding

**Method:** Methods: Patients with TLE (n = 52) and healthy volunteers (n = 35) completed a WM task, during which 34-channel electroencephalogram signals were recorded. The neural networks during WM encoding were calculated in TLE patients with (TLE-WM) and without (TLE-N) WM deficits

**Results:** Results: Functional connectivity strength decreased, and the theta network was altered in the TLE-WM group, although no significant differences in clinical features were observed between the TLE-N and TLE-WM groups

**Conclusion:** Conclusions: Not all patients with TLE present with cognitive impairments and alterations in the theta network were identified in TLE patients with functional cognitive deficits. Significance: The theta network may represent a sensitive measure of cognitive impairment and could predict cognitive outcomes among patients with TLE.

## 3. Cortical-subcortical neural networks for motor learning and storing sequence memory

Lanyun Cui, Ying Yu, L. Yin, Songan Hou, Qingyun Wang  
Neural Networks, 2025

Source: Semantic Scholar; matched terms: working memory, neural network, neural networks

Link: <https://www.semanticscholar.org/paper/77470db24128c991dc67873f76428cfd54dff287>

**Goal:** Motor sequence learning relies on the synergistic collaboration of multiple brain regions

**Method:** However, most existing models for motor sequence learning primarily focus on functional-level analyses of sequence memory mechanisms, providing limited neurophysiological insights into how biological neural systems intrinsically encode the ordering of sequential element

**Results:** These biologically inspired results aim to offer a mechanistic explanation for sequence learning and memory formation in the human brain, providing valuable insights into brain-like control systems and neural networks.

**Conclusion:** Finally, we demonstrate the model's applicability by deploying it to control a robotic arm in drawing and handwriting tasks, highlighting its adaptability to complex real-world scenarios These biologically inspired results aim to offer a mechanistic explanation for sequence learning and memory formation in the human brain, providing valuable insights into brain-like control systems and neural networks.

## 4. Geometry of neural computation unifies working memory and planning

Daniel B. Ehrlich, J. Murray  
bioRxiv, 2021

Source: Semantic Scholar; matched terms: working memory, neural network, neural networks, recurrent neural network

Link: <https://www.semanticscholar.org/paper/8be209193973b45110823bf969f1b6d8bdb1896d>

**Goal:** Real-world tasks require coordination of working memory, decision making, and planning, yet these cognitive functions have disproportionately been studied as independent modular processes in the brain

**Method:** Our experiments revealed that human behavior is consistent with contingency representations, and not with traditional sensory models of working memory

**Results:** Our experiments revealed that human behavior is consistent with contingency representations, and not with traditional sensory models of working memory

**Conclusion:** Finally, we generated falsifiable predictions for neural data to identify contingency representations in neural data and to dissociate different models of working memory Our findings characterize a neural representational strategy that can unify working memory, planning, and context-dependent decision making.

## 5. Reviewer #2 (Public review): Neural dynamics of visual working memory representation during sensory distraction

Authors unavailable  
2025

Source: Crossref; matched terms: visual working memory, working memory representation, working memory

Link: <https://doi.org/10.7554/elife.99290.2.sa1>

**Goal:** The title and metadata suggest relevance to Lila's working-memory and attention topics, but no abstract was available from the source API.

**Method:** Method details were not available in the retrieved metadata.

**Results:** Result details were not available in the retrieved metadata.

**Conclusion:** Open the linked record to inspect the full abstract or paper before journal-club use.